Unitary Method

If two persons X and Y can complete some work individually in a and b days, then their one day's work will be $\frac{1}{a}$ and $\frac{1}{b}$ respectively.

Working together, their one day's work $= \frac{1}{a}+\frac{1}{b} = \frac{a+b}{ab}$ and so,

X and Y together can complete the work in $\frac{ab}{a+b}$ days.

Example 4.23

A and B together can do a piece of work in 16 days and A alone can do it in 48 days. How long will B take to complete the work?

Solution:

$$(A+B)\text{'s 1 day's work} = \frac{1}{16}$$

$$\text{A's 1 day's work} = \frac{1}{48}$$

$$\therefore \text{ B's 1 day's work} = \frac{1}{16}-\frac{1}{48}$$

$$= \frac{3-1}{48} = \frac{2}{48} = \frac{1}{24}$$

$\therefore$ B alone can complete the work in 24 days.



Note

The time taken to complete a work or task depends on various factors such as **number of persons**, their **capacity** to do the work, the **amount of work** and the **time spent per day** for the completion of work.

Example 4.24

P and Q can do a piece of work in 20 days and 30 days respectively. They started the work together and Q left after some days of work and P finished the remaining work in 5 days. After how many days from the start did Q leave?

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Solution:

$$\text{P's 1 day's work} = \frac{1}{20} \text{ and Q's 1 day's work} = \frac{1}{30}$$

$$\text{P's work for 5 days} = \frac{1}{20}\times 5 = \frac{5}{20} = \frac{1}{4}$$

Therefore, the remaining work $= 1-\frac{1}{4} = \frac{3}{4}$ (as the total work is always 1). This remaining work was done by both P and Q.

Work done by P and Q in a day $= \frac{1}{20}+\frac{1}{30} = \frac{5}{60} = \frac{1}{12}$

Therefore, the number of days they worked together $= \frac{3/4}{1/12} = \frac{3}{4}\times\frac{12}{1} = 9$ days.

So, Q left after 9 days from the day the work started.

Example 4.25

A works 3 times as fast as B and is able to complete a work in 24 days less than the days taken by B. Find the time in which they can complete the work together.

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